

Psychrometric Chart Guide

What each feature does, and when you would reach for it

The Psychrometrics calculator (Calculators > Psychrometrics) plots air conditions and HVAC processes on a live psychrometric chart and does the load arithmetic for you. This guide walks through every feature with a worked, real-world example. All numbers in the examples were produced by the calculator itself, so you can re-enter them and see the same result.

How the screen is laid out

Settings strip (top of the left panel): Altitude, IP / SI units, **Save to project** and **Export PDF**.

Air Handler tab: Pick a system type, then add or remove components with the chips (Outdoor air, Return air, Cooling coil, Reheat, Space check ...). Each component gets its own section of inputs; check results appear at the bottom of the section they belong to.

State Points tab: Plot up to a handful of independent air conditions with no equipment attached. Use it to look up properties or compare conditions.

Results (below the inputs): Loads for every stage, condensate, and a full property table for every plotted point.

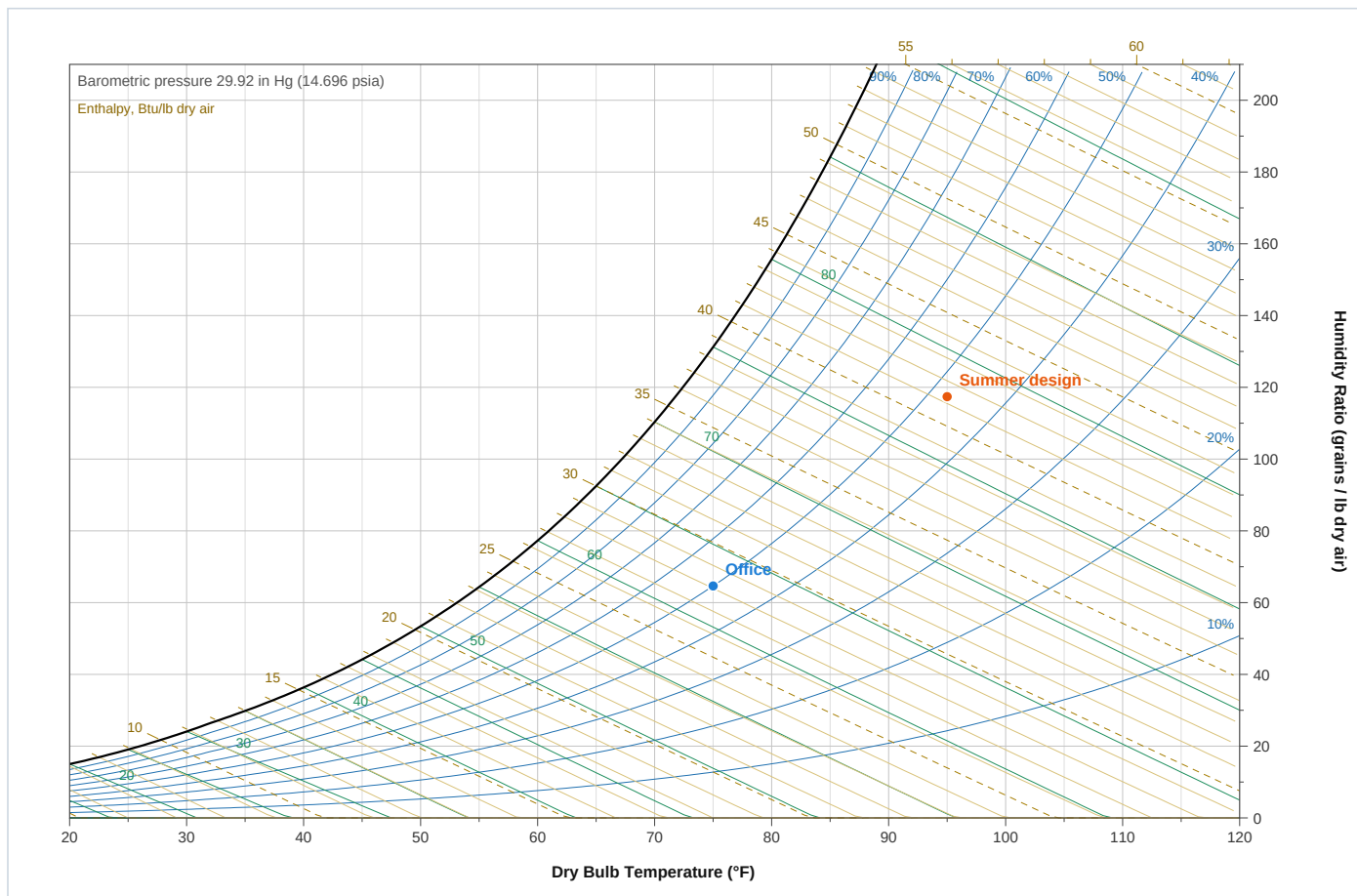
Chart toolbar: **Lines:** RH, Wet bulb, Enthalpy, Sp. volume. **Scales:** SHR protractor, SHF scale, Right-hand scales, and the reference point they use. Range, Fit and Reset control the view.

Chart: Wheel to zoom, drag to pan, hover to read the air state under the pointer in the bar below the chart.

What is in this guide

- 2 **Reading the chart, State Points** – Look up properties; estimate a ventilation load
- 3 **Mixed air + cooling coil** – Entering-coil condition and tonnage for a rooftop unit selection
- 4 **ADP / bypass factor, fan heat** – Draw-through units; what the coil must really leave
- 5 **Space check: supply air** – Will this supply air hold the room temperature *and* humidity?
- 6 **Space check: ventilation air only (DOAS)** – Required supply dew point for a DOAS serving VRF / chilled beams
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- 11 **Right-hand scales** – Dew point for condensation checks; vapor pressure
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- 13 **Saving, PDF export, quick reference** – Submittals and project records; which feature answers which question

Reading the chart



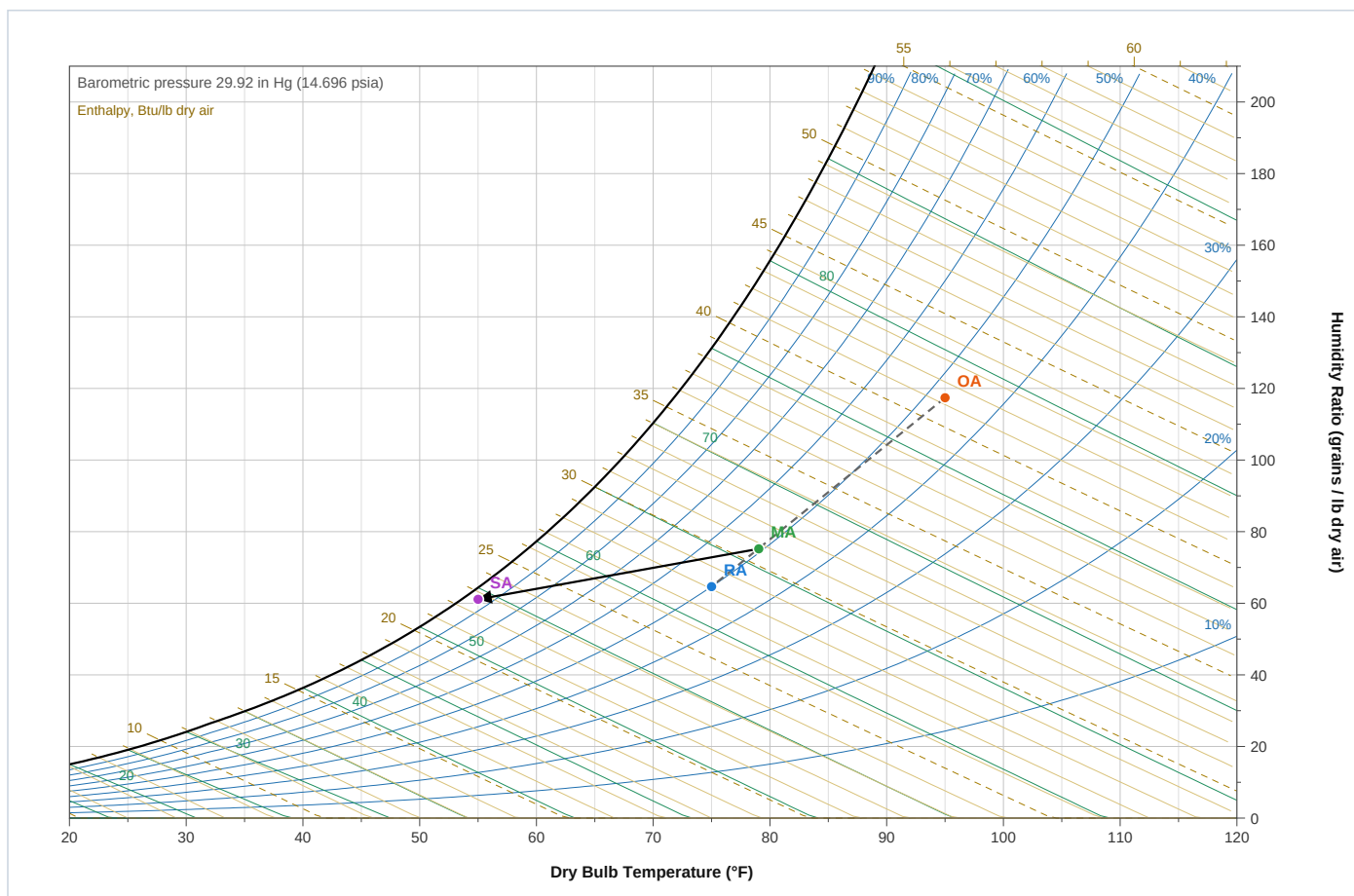
State Points tab: an office at 75 °F / 50% RH and a summer design day at 95 °F DB / 78 °F WB.

What you are looking at

- **Dry bulb** runs along the bottom, **humidity ratio** (grains of moisture per lb of dry air) up the right side. The heavy curve is **saturation** (100% RH); nothing exists above it.
- **RH** curves sweep up to the right (blue), **wet bulb** lines slope down to the right (green), **enthalpy** lines are the dashed amber family, and **Sp. volume** (off by default) adds the steep dotted lines.
- The **Enthalpy** checkbox also draws fine lines every 1 Btu/lb and a graduated scale just outside the saturation curve that continues along the top edge. Follow an enthalpy line up and to the left to read its value, exactly as on a printed chart.
- **Hover** anywhere under the curve and the bar below the chart reads out every property at that spot. **Wheel** zooms about the pointer, **drag** pans, **Fit** frames the plotted points, **Reset** returns to the standard view, and **Range** extends the chart down to cold-climate temperatures.

Real-world example: a quick ventilation load. An owner wants to add 1,000 CFM of outdoor air to an existing office system and asks what it will cost in cooling. On the State Points tab enter the room (75 °F, 50% RH) and the summer design day (95 °F DB, 78 °F WB). The property table gives enthalpies of **28.11** and **41.30 Btu/lb**. The difference, 13.19 Btu/lb, times $4.5 \times 1,000$ CFM is about **59,400 Btu/h, or 5 tons**, before you have opened any selection software. The same table gives the room dew point (55.1 °F), useful for the condensation checks on page 11.

Mixed air and the cooling coil



System type "Packaged RTU / split (mixed air)": 2,000 CFM outdoor air + 8,000 CFM return air, coil leaving 55 °F / 95% RH.

What it does

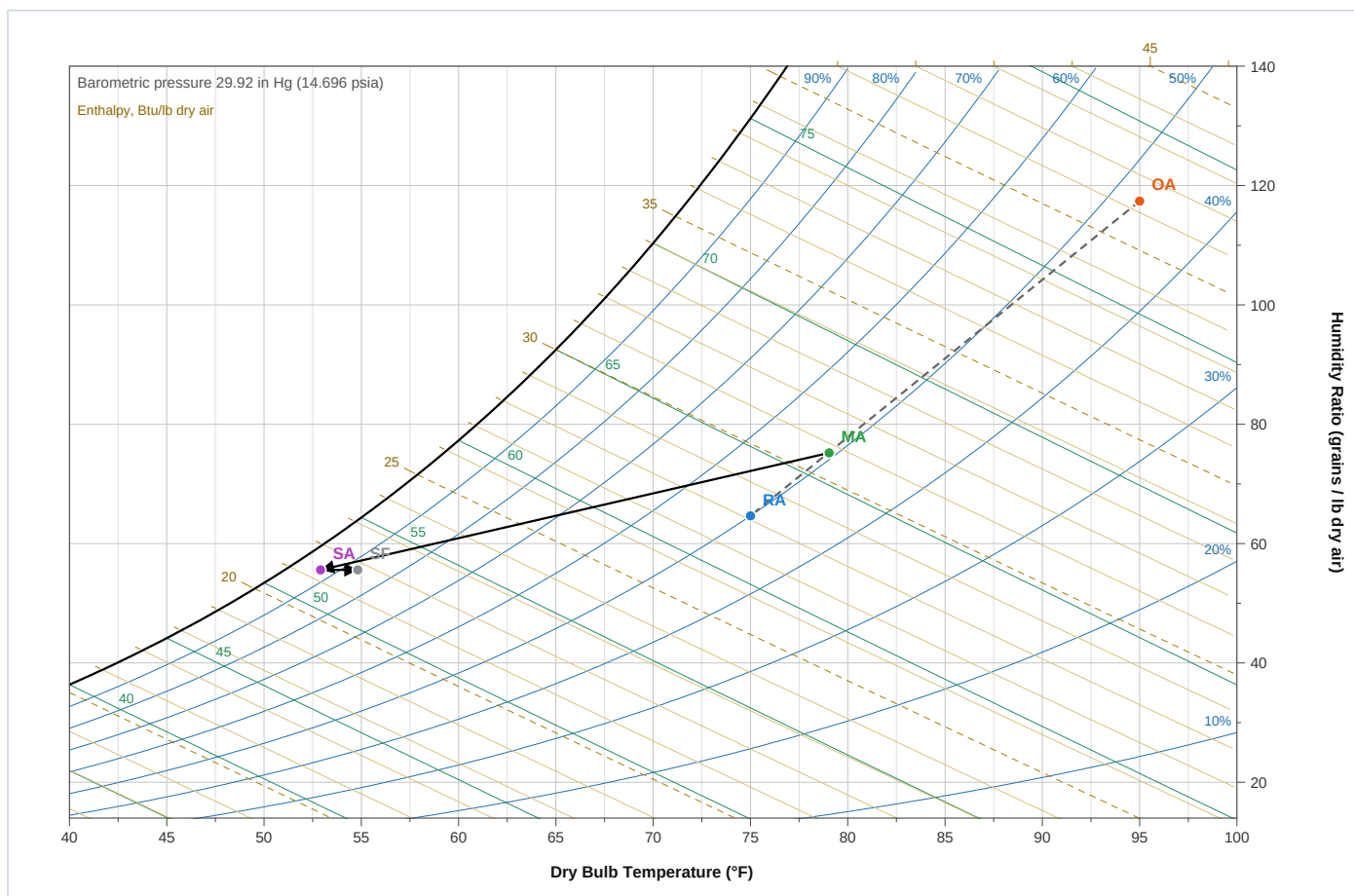
Outdoor air (OA) and return air (RA) are joined by the grey dashed mixing line; the mixed air point (MA) sits on that line in proportion to the airflows. The solid arrow is the cooling coil process from MA to the supply air (SA). The Results list the coil total, sensible and latent load, its sensible heat ratio, the apparatus dew point and bypass factor the process implies, the moisture removed, condensate flow and the code-minimum drain size.

Airflow can be entered per stream, or as a total with a percentage of outdoor air. If you already know the entering-coil condition, use "Mixed air (known)" instead of OA and RA.

Real-world example: getting the numbers a rooftop unit selection needs. A 10,000 CFM unit serves a retail space with 20% outdoor air. Design day is 95 / 78 °F, return air is 75 °F / 50% RH. The calculator puts the entering-coil condition at **79.0 °F DB / 66.1 °F WB**, which is the pair the selection software asks for. Leaving the coil at 55 °F / 95% RH takes **363,200 Btu/h (30.3 tons)**: 263,900 sensible and 99,300 latent, an SHR of 0.73. The coil wrings out 90.6 lb/h of water, **10.9 gal/h** of condensate, and the drain needs to be at least 1 in. (IMC 307.2.2). You now know to look at 30-ton units, and you have a condensate figure for the plumbing engineer.

Tip: the report shows "Apparatus dew point 53.1 °F, bypass factor 7.3%". If a manufacturer's coil cannot reach that ADP with your chilled-water or refrigerant temperature, the leaving condition you asked for is not achievable.

Coil by ADP and bypass factor; fan heat



Same unit, coil described by a 50 °F apparatus dew point and 10% bypass factor, followed by a 7.5 bhp draw-through supply fan (SF).

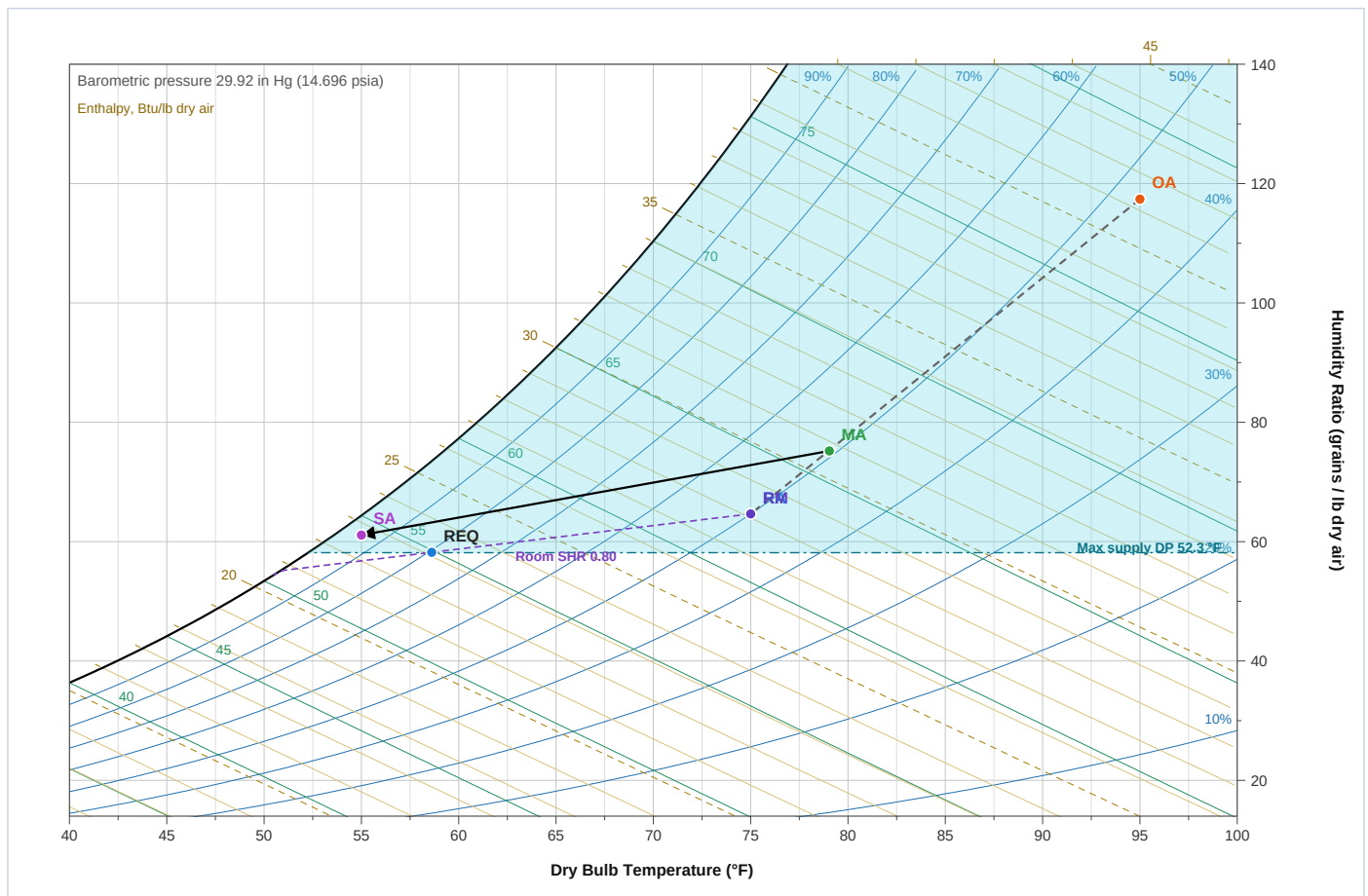
What it does

- **Define by: ADP + bypass factor.** Instead of stating the leaving air, describe the coil: the saturation temperature its surface runs at and the fraction of air that slips past untouched. The calculator finds the leaving point on the straight line from the entering air to the ADP.
- **Fan heat** adds the supply fan (and motor, if it sits in the airstream) as a short horizontal step to the right. Enter brake horsepower or a temperature rise.

Real-world example: why the space was 2 °F warm at start-up. A draw-through air handler was selected for 55 °F supply air, but the balancer measures 57 °F at the first diffuser. With the coil at ADP 50 °F / 10% bypass the air leaves the coil at **52.9 °F**. A 7.5 bhp fan with a 91% motor in the airstream adds **21,000 Btu/h, a 1.9 °F rise**, so the unit actually delivers **54.8 °F**. Net cooling to the space is 403,700 Btu/h, not the 424,600 Btu/h the coil is doing. If the coil had been selected to *leave* 55 °F, the supply would be near 57 °F, which is exactly what was measured: the coil needs to be selected about 2 °F colder.

Use ADP + bypass when you are checking an existing coil or comparing row counts (4 to 6 row coils typically run 5 to 15% bypass). Use Leaving condition when you know what you want and need to see what coil that implies.

Space check: supply air carries the load



Latent load carried by: Supply air. Room 75 °F / 50% RH with 180,000 Btu/h sensible and 45,000 Btu/h latent. Chart zoomed with the mouse wheel.

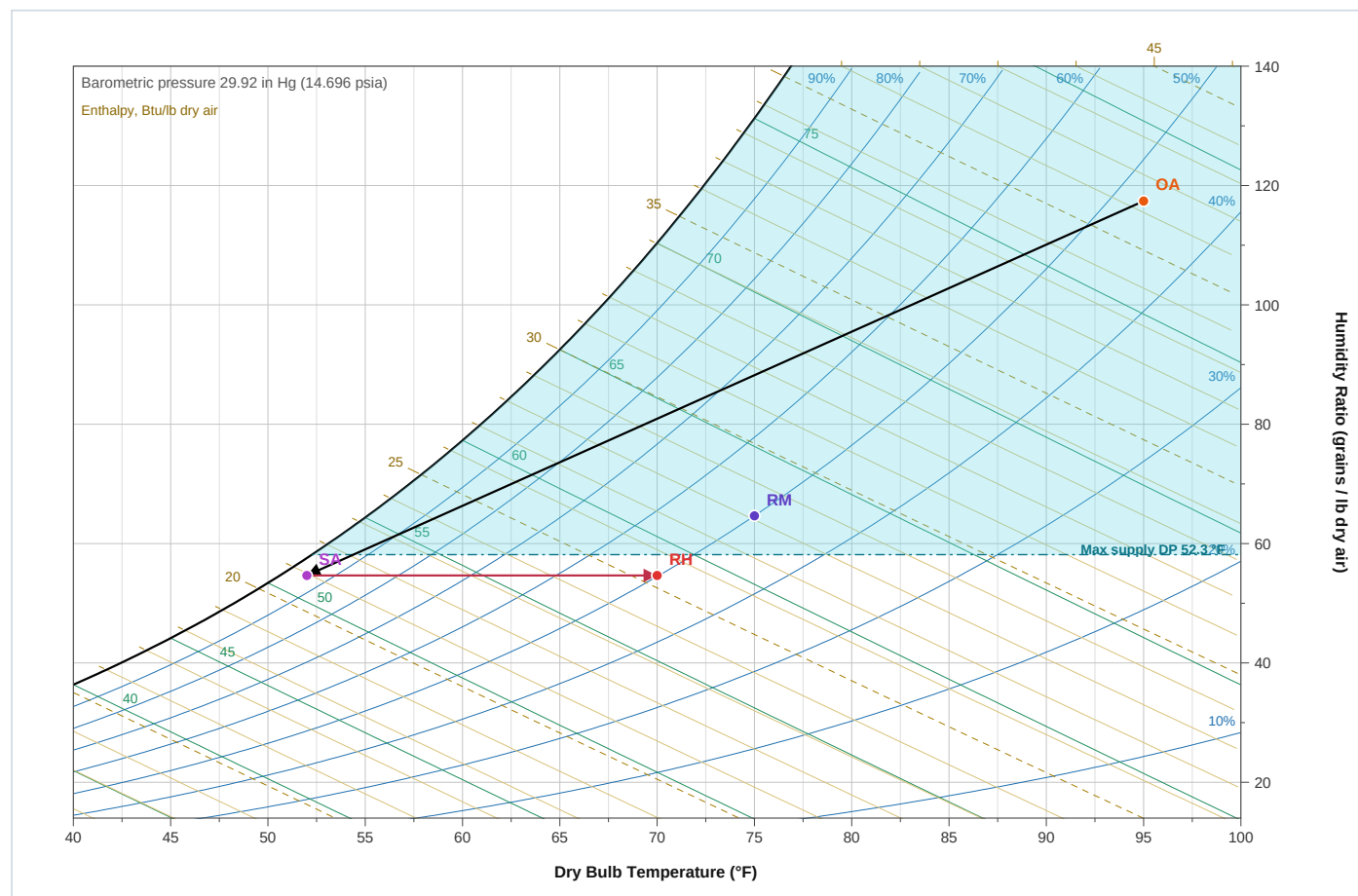
What it does

Enter the space condition and its sensible and latent loads. The calculator draws the dashed purple **room SHR line**: any supply air on this line removes heat and moisture in the same proportion the room produces them. **REQ** marks where your airflow lands on that line, the supply condition that would exactly carry the load. The cyan line and shaded band show the **maximum supply dew point**; supply air inside the band is too humid. Results appear under the Space section. A blank Supply airflow uses the system airflow (outdoor + return), and a note under the field shows the value in use.

Real-world example: “the office is cool but clammy.” An open office has 180,000 Btu/h sensible and 45,000 Btu/h latent (SHR 0.80) on a 10,000 CFM unit leaving 55 °F / 95% RH. The check shows REQ at **58.6 °F with a 52.3 °F dew point**. The unit’s supply air is colder than needed (sensible 219,500 of 180,000 Btu/h, OK) but its dew point is **53.6 °F, above the 52.3 °F maximum**: it only removes about 25,000 of the 45,000 Btu/h latent load. On the chart SA sits inside the shaded band and above the room line. The space will hold temperature and drift above 50% RH. The fix is a lower coil leaving dew point (then reheat or reset the airflow), not more tonnage.

Solve for “Airflow from supply temperature” when the supply temperature is fixed (for example 55 °F) and you need the CFM. If carrying the load would take supply air below 40 °F, the result is reported as not practical and REQ is not plotted: raise the airflow or add return air.

Space check: ventilation air only (DOAS)



System type "100% outdoor air / DOAS": 2,000 CFM, coil leaving 52 °F / 95% RH, reheat to 70 °F. Latent load carried by: Ventilation air only.

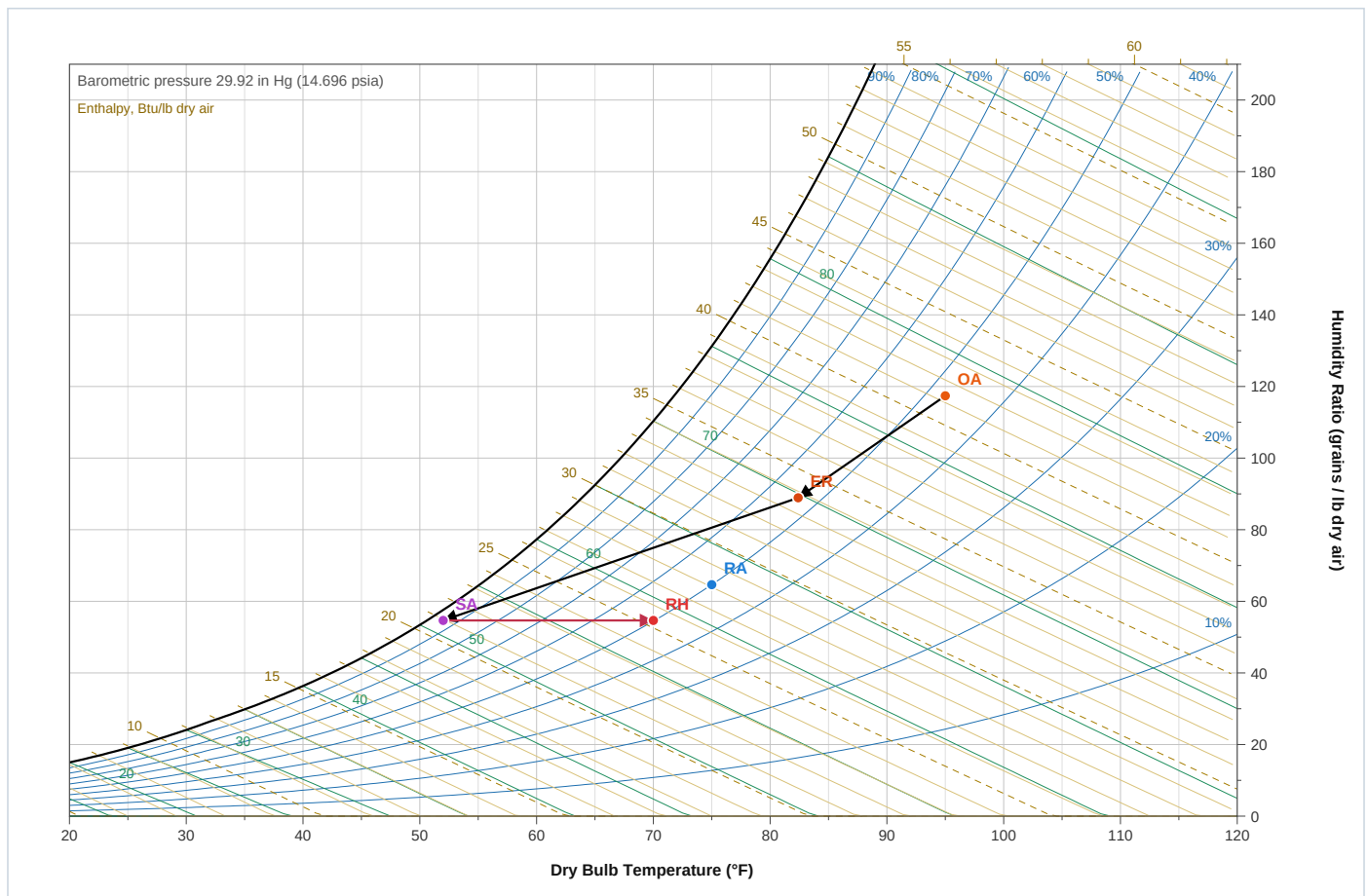
What it does

With a dedicated outdoor air system the zone equipment (VRF, chilled beams, fan coils) is meant to run dry, so the ventilation air is the only thing removing moisture from the space. Choose **Ventilation air only (DOAS)** and the Space section asks for just the space condition, the latent load, the ventilation airflow (blank = the outdoor airflow) and, optionally, the unit's rated supply dew point. It returns the **maximum supply dew point**: $Q_{\text{latent}} = 0.69 \times \text{CFM} \times \Delta \text{gr/lb}$. The sensible load is left to the zone units.

Real-world example: DOAS for a VRF classroom wing. 45 occupants at 200 Btu/h each is a 9,000 Btu/h latent load; ventilation is 2,000 CFM; the rooms are to hold 75 °F / 50% RH (64.6 gr/lb). The air must be delivered 6.5 gr/lb drier than the room, so the **maximum supply dew point is 52.3 °F**. The unit modelled here leaves the coil at a 50.6 °F dew point: **OK**, carrying 13,800 Btu/h against the 9,000 needed. A unit rated for a 50 °F supply dew point also passes, and would still work down to 1,160 CFM. Try a unit that only reaches 55 °F and the check turns red and tells you the airflow it would need instead.

The reheat step (SA to RH) shows the neutral-air delivery at 70 °F: the dew point does not change, only the dry bulb, and the 39,400 Btu/h of reheat is reported separately. The Supply air dew point calculator on the Calculators page hands its inputs to this same check.

Energy recovery



DOAS with a wheel: 3,000 CFM outdoor air, 2,700 CFM exhaust at the return air condition, 70% sensible / 60% latent effectiveness.

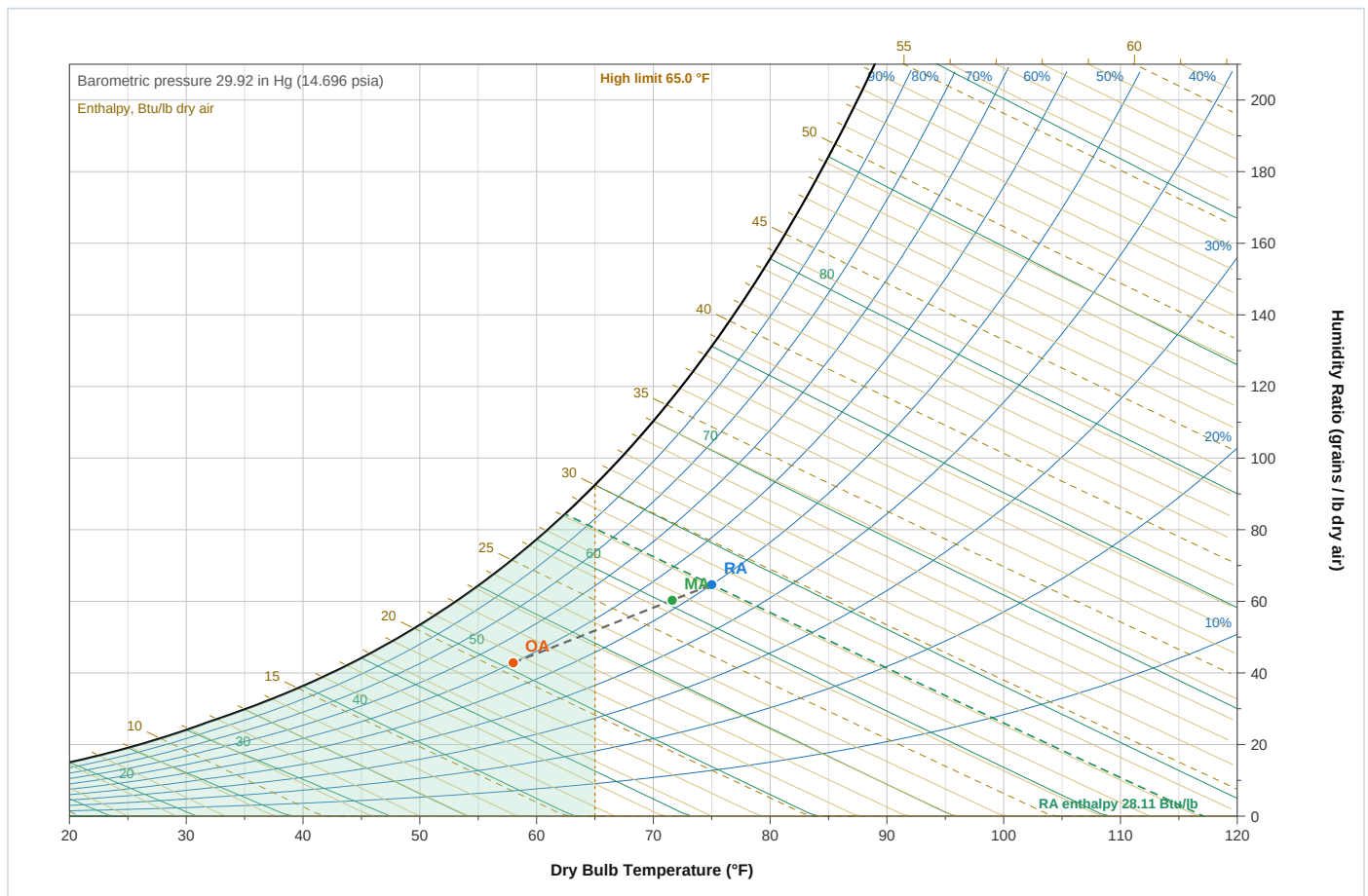
What it does

Add **Energy recovery** (it needs both Outdoor air and Return air) and the outdoor air is pre-conditioned toward the return air condition; the leaving point plots as **ER**. Sensible effectiveness moves ER horizontally, latent effectiveness vertically (plates have little or none). Two arrangements: *return air mixes* after the device, or *dedicated outdoor air unit*, where return air is only the exhaust side and the coil sees ER directly. If the exhaust flow is smaller than the outdoor flow, recovery is scaled down to match.

Real-world example: is the wheel worth it? A 3,000 CFM DOAS on a 95 / 78 °F design day must deliver a 50.6 °F dew point. Without recovery the coil would see raw outdoor air: $4.5 \times 3,000 \times (41.30 - 20.94)$ is about 274,900 Btu/h, **22.9 tons**. With a 70 / 60% wheel and 2,700 CFM of exhaust, the outdoor air reaches the coil at **82.4 °F DB / 69.8 °F WB**; the wheel has removed 102,400 Btu/h (8.5 tons) and the coil load falls to **172,400 Btu/h, 14.4 tons**. That is the difference between a 25-ton and a 15-ton unit, and the ER condition is the entering-coil condition to give the manufacturer.

Run the same model with winter conditions to see the preheat the wheel saves, and to check whether ER approaches saturation (frost risk on the exhaust side).

Economizer check



Differential enthalpy changeover with a 65 °F high limit. Outdoor air 58 °F / 60% RH; return air 75 °F / 50% RH.

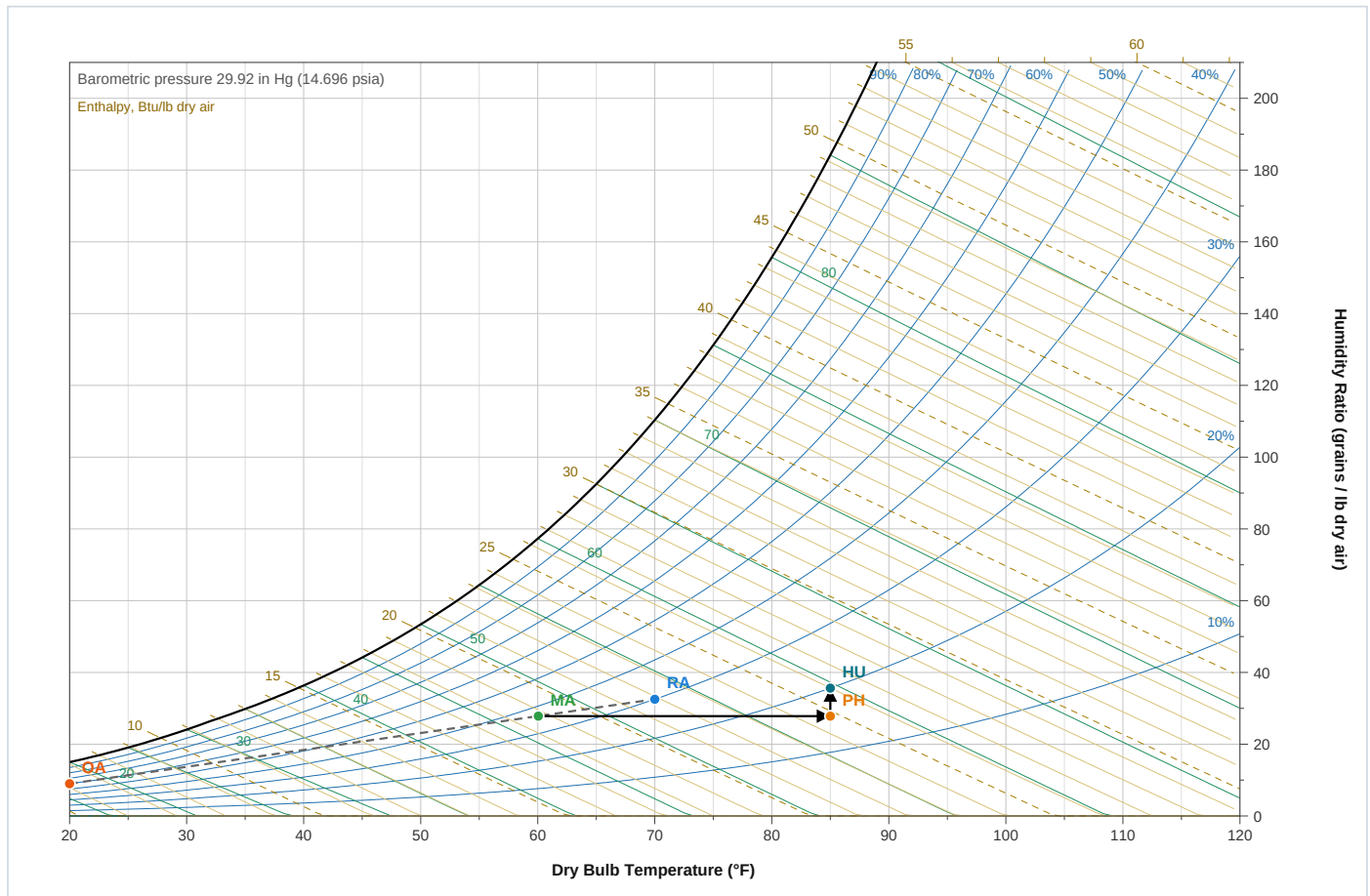
What it does

Compares outdoor air with return air and tells you whether free cooling applies. The green dashed line is the changeover boundary (the return-air enthalpy line for differential enthalpy, or a vertical line at the return dry bulb for differential dry bulb), the amber dotted line is the fixed high limit, and the shaded area is where outdoor air must fall for the dampers to open. The verdict is listed under the Economizer section.

Real-world example: checking controls settings during commissioning. On a 58 °F, 60% RH morning the outdoor air enthalpy is 20.57 Btu/lb against 28.11 for the return air: **7.5 Btu/lb lower, and 7 °F under the 65 °F high limit, so free cooling is available** and OA plots inside the shaded region. Now enter a muggy 68 °F / 90% RH afternoon: it is cooler than the return air, so a dry-bulb changeover would open the dampers, but its enthalpy is higher than the return air and the enthalpy changeover correctly keeps them at minimum. Switching the Changeover option between the two modes shows the owner, on one chart, why the enthalpy sensor is worth having in a humid climate.

The economizer needs both Outdoor air and Return air in the system; its chip is greyed out otherwise.

Preheat and humidification



System type "Winter humidification": 20 °F outdoor air, 70 °F / 30% RH return air, preheat to 85 °F, steam humidifier.

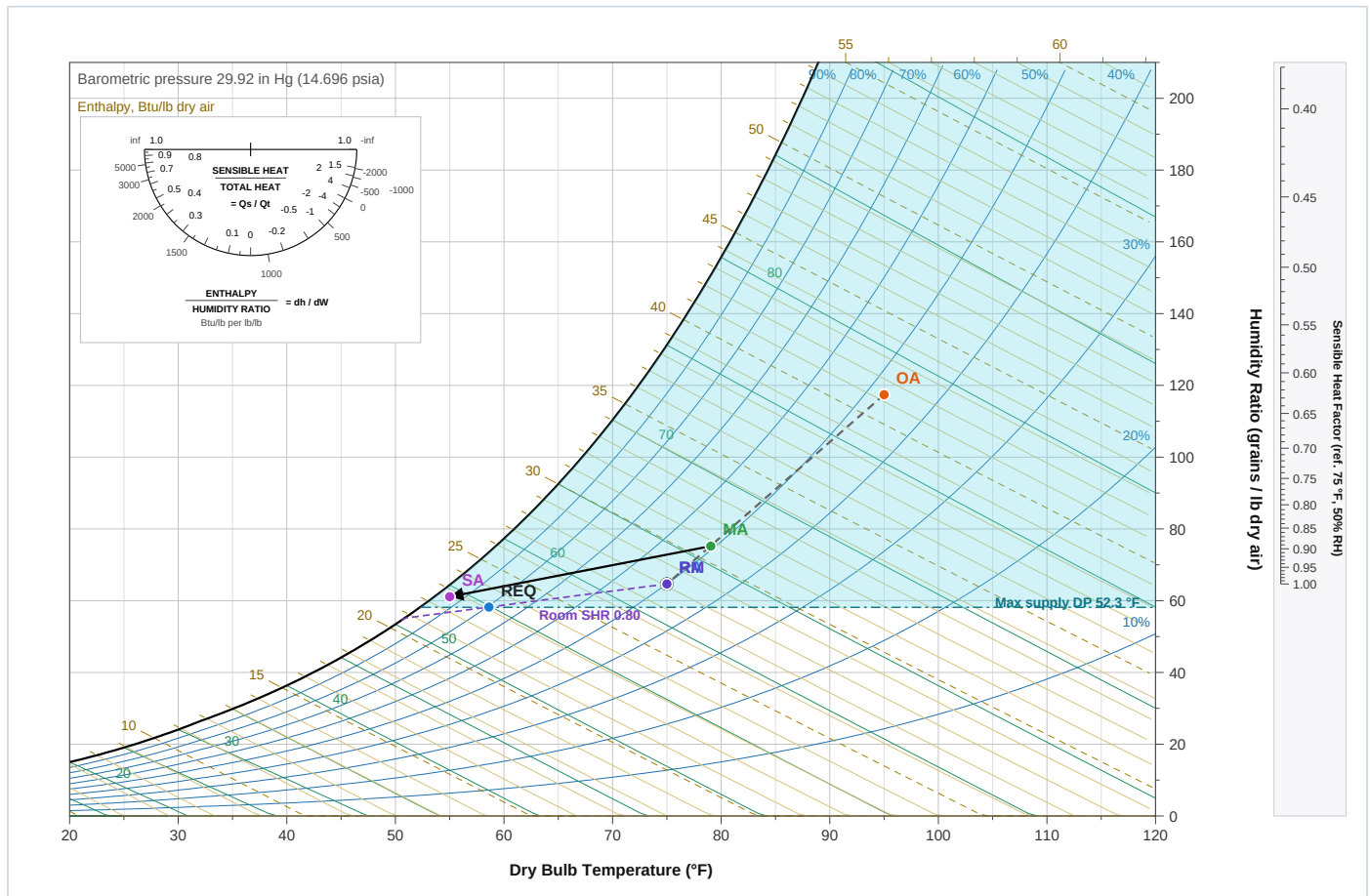
What it does

- **Preheat / heating coil** is sensible heating: a horizontal move to the right at constant moisture (PH). Note how the RH collapses as the air warms.
- **Humidifier:** *steam* rises almost vertically to a target RH, dew point or humidity ratio; *evaporative* moves up and to the left along the wet-bulb line by its effectiveness, cooling the air as it humidifies. The Results give the water added and, for steam, the heat that comes with it.

Real-world example: sizing a steam humidifier for a print shop. The space must not fall below 30% RH at 70 °F (32.5 gr/lb) or paper curls and static builds. On a 20 °F day, 2,000 CFM of outdoor air at 9 gr/lb mixes with 8,000 CFM of return air to give 60 °F / 27.8 gr/lb. Preheating to 85 °F takes **271,400 Btu/h** and leaves the air at only 15.6% RH. To deliver supply air slightly wetter than the room (35.6 gr/lb, enough to offset dry infiltration), the humidifier must add **50 lb/h of steam**, which also adds 55,200 Btu/h of heat. That 50 lb/h is the number the humidifier manufacturer needs; repeat with your coldest, driest design day.

Check the humidified point against saturation: if HU lands close to the curve, moisture will condense in the duct downstream of the dispersion grid.

SHR protractor, SHF scale and the reference point



SHR protractor and SHF scale switched on, reference point 75 °F / 50% RH (the small ring). The room line from page 5 is also shown.

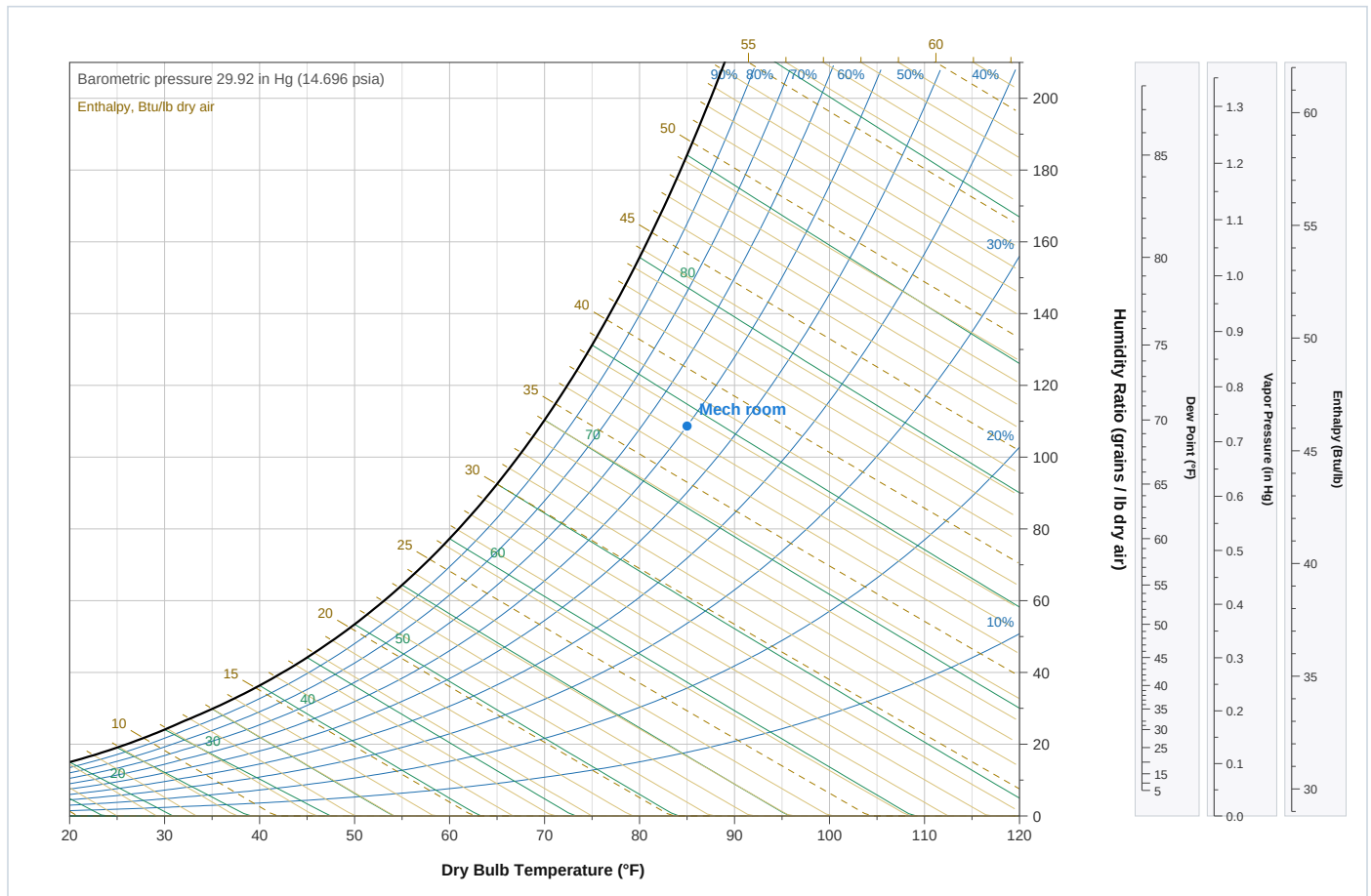
What they are

- **SHR protractor** (top left). Inner scale: sensible heat / total heat. Outer scale: enthalpy change per unit of humidity ratio change. A process line on the chart has the same slope as the matching line from the protractor's centre to that value.
- **SHF scale** (right). A straight line from the **reference point** ring to a value on this scale has that sensible heat factor's slope.
- **Ref. point.** Defaults to 75 °F / 50% RH like most printed charts. Change it when your room design condition is different, so the scale is exact for lines through your room.

Real-world example: a sanity check before the loads are entered. A load program reports a room SHR of 0.80. With the scales on, lay a straightedge from the ring at 75 / 50% to 0.80 on the SHF scale and follow it down-left to the saturation curve: it lands near 51 °F, so you know at a glance this room needs a coil with an apparatus dew point of about 51 °F or lower, and that a unit with a 0.90 SHR coil will not hold humidity. When you later enter the loads in the Space section, the dashed room line drawn by the calculator falls on the same slope. For a lab designed to 72 °F / 45% RH, type 72 and 45 into Ref. point and the ring, protractor and scale all redraw for that condition.

These scales are optional and off by default to keep the chart clear. The calculator's own room line is exact; the scales are the traditional graphical method and a good cross-check. The plot narrows slightly while right-hand scales are shown.

Right-hand scales



Right-hand scales on: dew point, vapor pressure and enthalpy columns, each in its own box. Point: mechanical room at 85 °F / 60% RH.

What they are

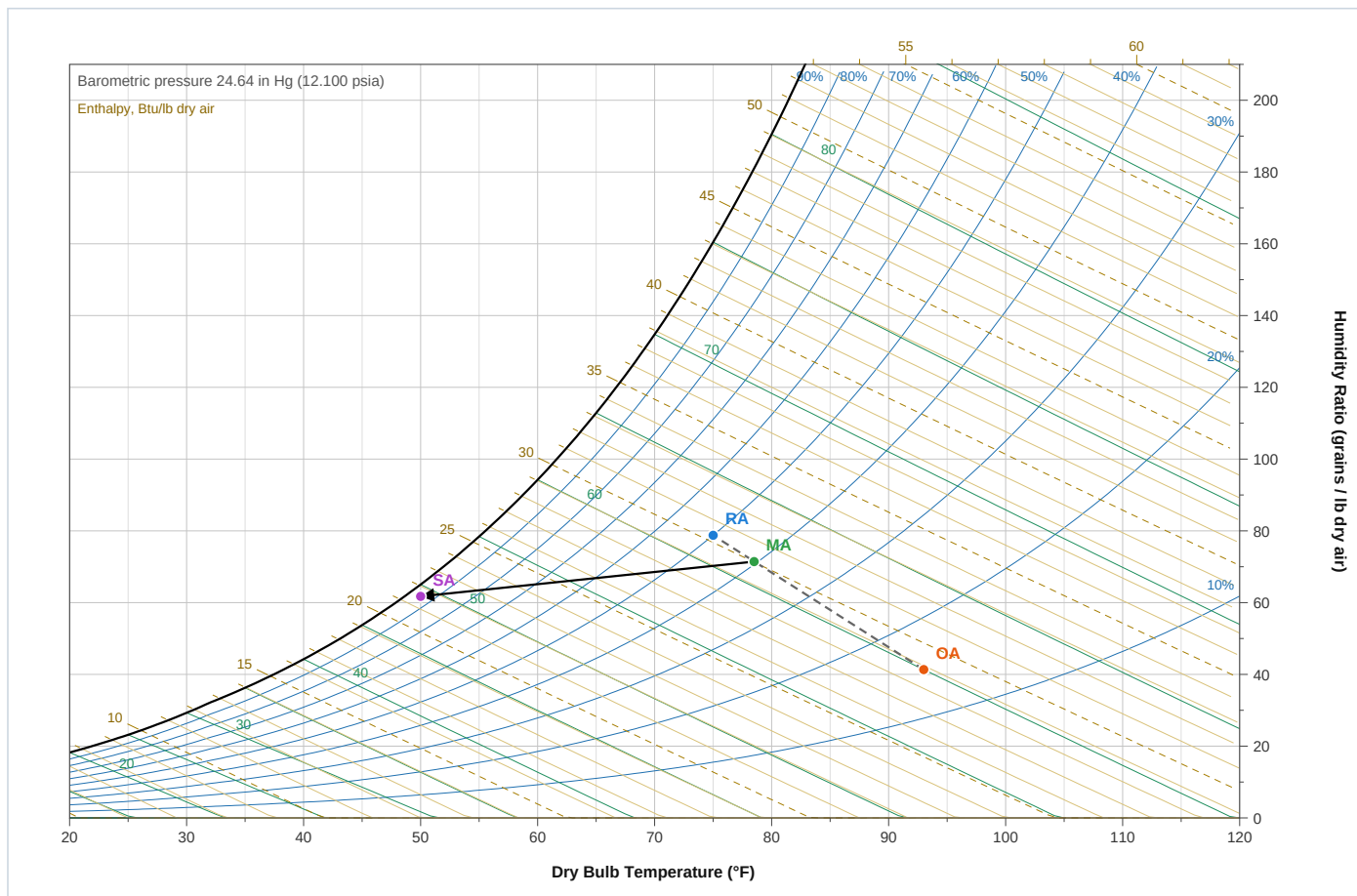
Three extra columns to the right of the humidity ratio axis, as on a printed chart. Carry any point horizontally to the right:

- **Dew point:** the surface temperature at which that air starts to condense.
- **Vapor pressure:** the partial pressure of the water vapor, the driving force for moisture migration through walls and insulation.
- **Enthalpy:** marks where each enthalpy line meets the right-hand edge of the chart; follow an enthalpy line down to the right to read it.

Real-world example: will the chilled water piping sweat? A mechanical room sits at 85 °F and 60% RH in summer. Plot it on the State Points tab, switch on Right-hand scales, and read straight across: the **dew point is about 70 °F** (69.6 °F in the table). Chilled water at 45 °F is far below that, so any bare fitting or gap in the vapor barrier will drip. The vapor pressure reads about **0.73 in. Hg**; inside a 38 °F cooler it is roughly 0.2 in. Hg, and that difference is what drives moisture into cooler walls and pipe insulation, which is why the vapor barrier goes on the warm side.

The same horizontal reading works for supply air: a 55 °F / 95% RH supply has a 53.6 °F dew point, so uninsulated supply duct in a 75 °F / 60% RH ceiling plenum (60 °F dew point) will sweat.

Altitude, units and load basis



The mixed-air unit at 5,280 ft (24.9 in. Hg): 93 °F / 60 °F WB outdoor air, coil leaving 50 °F / 95% RH, actual-air basis.

What they do

- **Altitude** sets the barometric pressure used for every property. Higher up, the saturation curve rises and the same RH holds more moisture; the chart redraws for that pressure.
- **IP / SI** switches every input, axis, scale and result. The maths does not change.
- **Load basis** (Airflow section). *Standard air* uses 0.075 lb/ft³ (CFM × 4.5), the convention in coil selection software and the 1.08 / 0.68 rules of thumb. *Actual air* uses the real specific volume of each stream, which matters at altitude or high temperature. It changes the loads, not the plotted points.

Real-world example: a rooftop unit in Denver. At sea level a 75 °F / 50% RH room holds 64.6 gr/lb; at 5,280 ft the same room holds **78.8 gr/lb**, so grains-based rules from a sea-level chart are simply wrong. More important, 10,000 CFM is 45,000 lb/h of air at sea level but only **35,800 lb/h** in Denver, about 20% less air to carry heat. The coil in the figure does 303,400 Btu/h (25.3 tons); worked with standard air the same airflow would appear to do about 25% more. Set the altitude first, use the actual-air basis, and tell the manufacturer the elevation so their rating is corrected the same way.

The pressure in use is printed in the chart's top-left corner and next to the Altitude box, so every exported PDF records it.

Saving, exporting and quick reference

Saving and exporting

- **Save to project** stores the whole calculation (inputs, view and chart options) with the active project. It appears in the project view, where it can be reopened or exported later.
- **Export PDF** produces a report with the chart exactly as it is on screen, including whichever lines and scales are switched on and the current zoom, followed by every load, check result and the full property table. Zoom in on the process before exporting if the points are crowded.
- Checkbox states and the reference point reset when the page is reloaded; saved calculations keep theirs.
- **ASHRAE design conditions** (top right) opens the climatic design data site in a new tab to look up the design dry bulb / wet bulb for a weather station.

Real-world example: backing up a submittal. A reviewer questions why a DOAS was selected at a 50 °F supply dew point. Open the saved calculation, zoom the chart onto the OA to SA to RH process, and export. The PDF shows the chart with the maximum-dew-point line, the “Supply air dew point (Law #1)” block with the latent load, airflow and the 52.3 °F requirement, and the property table for every point: a short report with the arithmetic visible.

Which feature do I need?

Look up the properties of one or more air conditions

→ **State Points tab; hover readout**

Find the entering-coil condition and tonnage for a mixed-air unit

→ **Outdoor air + Return air + Cooling coil**

Check an existing coil, or see what ADP a leaving condition implies

→ **Cooling coil: ADP + bypass factor / reported ADP**

Account for a draw-through fan

→ **Fan heat**

See whether supply air will hold room temperature and humidity

→ **Space check: Supply air**

Find the supply dew point a DOAS must deliver

→ **Space check: Ventilation air only (DOAS)**

Show overcooling and reheat for dehumidification

→ **Cooling coil + Reheat**

Size the saving from a wheel or plate

→ **Energy recovery**

Verify economizer changeover and high limit

→ **Economizer check**

Size preheat or a humidifier

→ **Preheat / heating coil, Humidifier**

Work an SHR graphically, as on a paper chart

→ **SHR protractor, SHF scale, Ref. point**

Read dew point or vapor pressure for a condensation question

→ **Right-hand scales (or the property table)**

Work a project above sea level

→ **Altitude + Actual air basis**

Every input has a “?” next to it that explains what it is and how changing it moves the chart. Psychrometric properties follow ASHRAE Fundamentals (PsychroLib).